Sarbajit Ghosh

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Experience

Winter Internship – India Space Lab Feb 2026 – March 2026

- Designed and implemented a dual redundant PCB architecture for a CanSat system to ensure high reliability and fault tolerance in mission-critical aerospace applications.
- Developed the complete circuit schematic and 2-layer PCB layout, focusing on efficient component placement, optimized routing, and stable power distribution.
- Generated and validated Gerber files along with 3D PCB visualization, ensuring the design is fabrication-ready and mechanically compatible.
- Created the CAD model of the CanSat structure and explored embedded systems and microcontroller interfacing, enabling seamless integration of hardware components

Projects

EDURESQ | *ESP32, RFID, IoT, GPS NEO-8M, Web Server Integration*

- Developed an IoT-based system using RFID-enabled ID cards for automated exam seat allocation and hospital bed management.
- Designed embedded hardware architecture using ESP32 for real-time identification and data communication.
- Integrated GPS NEO-8M module for tracking and location-based resource allocation.
- Implemented web server integration for secure storage and retrieval of user and medical records.
- Reduced manual intervention and allocation errors through automated data processing.

Wearable Health Monitoring System | *Machine Learning, IoT, ESP32, Python, Embedded Systems, Data Analysis*

- Developed a wearable system for real-time monitoring of health parameters such as heart rate and body temperature.
- Integrated IoT modules for continuous data transmission and remote monitoring.
- Applied machine learning techniques to analyze health data and detect abnormal patterns.
- Designed embedded hardware for real-time data acquisition and processing.
- Improved monitoring reliability through sensor calibration and data analysis.

Skills

Programming: C++, Python
Libraries: Pandas, NumPy, Scikit-learn,Matplotlib
Tools: Xilinx, Cadence, TCAD, Arduino IDE, Fritzing, EasyEda, Git, VS Code,Jupyter Notebook
Embedded Systems and IoT:Arduino , ESP32, Raspberry Pi, PCB Designing and Fabrication, Sensor Integration

Education

Birla Institute of Technology Mesra, Ranchi 2024 – 2028
Bachelor of Technology : Electronics and Communication Engineering : SGPA: 8.61

Manisha International School, Durgapur, West Bengal 2022 – 2024
Higher Secondary (CBSE) : 79.4%

West End High School, Jhargram, West Bengal 2020 – 2022
Secondary (ICSE) : 94%

Accomplishments

- Won 1st Position at IETE Students' Innovation Day for EDU-RESQ, an RFID-based smart resource allocation system.
- Secured 2nd Position at BIT Mesra Internal Hackathon for innovative project development and implementation.

Volunteering Experience

- Coordinator at AI Club, BIT Mesra (Off-Campus Jaipur), organized technical sessions and collaborative learning activities on AI and emerging technologies.
- Coordinator at National Service Scheme (NSS), actively involved in planning and executing community service and social outreach activities.